Zhaofeng Sun

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Research interests: Language model architectures, machine learning systems, and model compression.

Education

Tsinghua University Sep 2022 - Jun 2027 (expected)
B.S. in Computer Science | GPA: 3.97/4.00 | Rank: 5/208

Cornell University Jan 2025 - Jun 2025
Exchange Student in Computer Science | GPA: 4.15/4.00

Research Experience

University of California San Diego Jun 2026 - Present
Prof. Hao Zhang · Hao AI Lab

- **Efficient Looped Architecture.** Co-designed a parallel recurrent architecture that improves training stability and depth scaling while retaining efficient inference. The eight-loop model improves average downstream accuracy by 0.84 percentage points over vanilla PLT with only 0.3% additional decoding time.

Carnegie Mellon University May 2025 - Aug 2025
Prof. Beidi Chen

- **Sparse Attention.** Co-designed sparse-attention rollout methods for long-context reinforcement learning, controlling actor-policy mismatch to sustain stable learning with up to 2.4× faster rollouts.

Cornell University Jan 2025 - May 2025
Prof. Christopher De Sa

- **Model-Preserving Adaptive Rounding.** Preserves model behavior after low-bit quantization by targeting end-to-end output error, with theoretical guarantees and approximately 30% lower KL divergence than GPTQ/LDLQ.

Tsinghua University Oct 2023 - Jan 2025
Prof. Jianfei Chen

- **Efficient model compression.** Hardware-aware sparsity and quantization connect compression choices to accelerator execution, aiming to turn model compression into practical inference speedups.

Selected Publications

Model-Preserving Adaptive Rounding
Albert Tseng, **Zhaofeng Sun**, Christopher De Sa.
ICML 2026 Paper

Sparrow: Sparse Rollout for Stable and Efficient Long-context RL of Large Language Models
Yang Zhou, Ranajoy Sadhukhan, **Zhaofeng Sun**, Zhuoming Chen, Souvik Kundu, Saket Dingliwal, Sai Muralidhar Jayanthi, Aram Galstyan, Haizhong Zheng, Beidi Chen.
Preprint, 2026 · Earlier version at SPOT, ICLR 2026 Paper

Beyond 2:4: Exploring V:N:M Sparsity for Efficient Transformer Inference on GPUs
Kang Zhao, Tao Yuan, Han Bao, Zhenfeng Su, Chang Gao, **Zhaofeng Sun**, Zichen Liang, Liping Jing, Jianfei Chen.
Preprint, 2024 Paper

Honors and Technical Skills

Champion, Tsinghua University Supercomputer Competition (2024).
Comprehensive Excellence Scholarship, top 5%, Tsinghua University (2023-2025).
Languages: Python, C/C++, CUDA, Triton. **Systems:** PyTorch, FSDP, Megatron, vLLM, SGLang, verl.